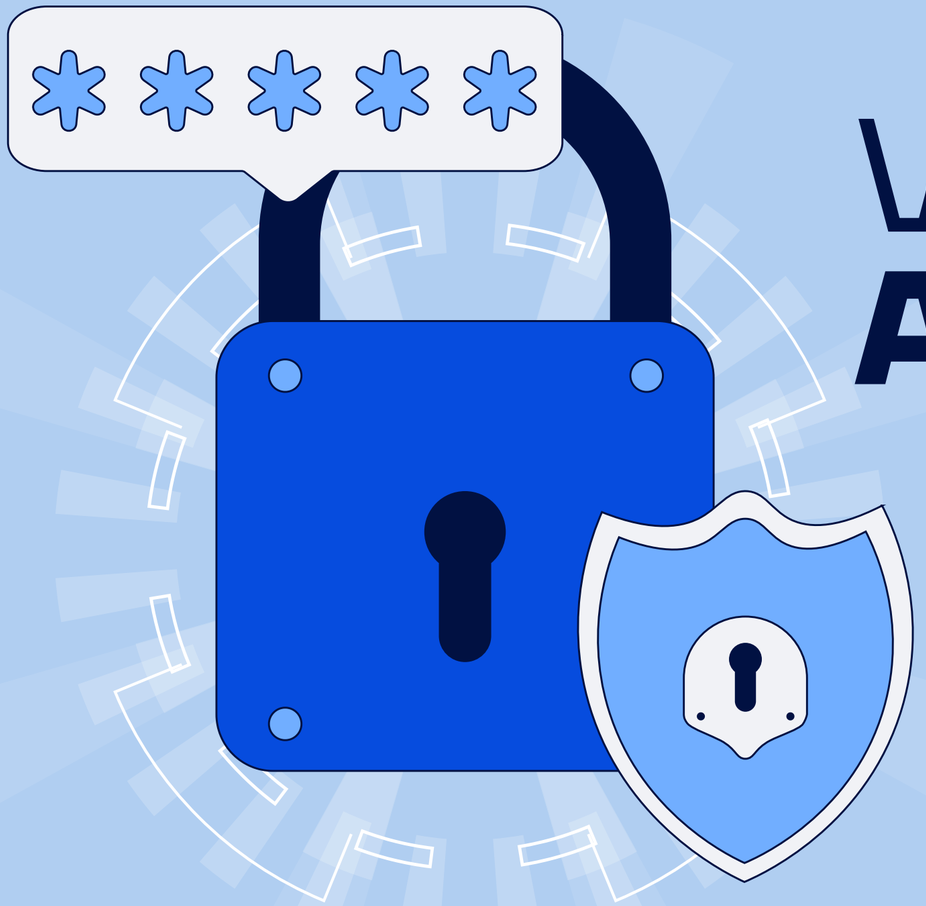


Vulnerability **Assessment Report**

Cyber Security Internship – Task 01

Prepared by: Afnan Muhammed
Track: Cyber Security (CS)

What I have **Analyzed**

- This report presents the results of a vulnerability assessment conducted on a publicly accessible website using industry-standard security tools.
- The objective of the assessment was to identify potential security weaknesses and misconfigurations that could expose the application to security risks.
- The assessment used passive scanning techniques to safely analyze the target system without causing disruption to the service.

Scope of **Assessment**

Target Website

Target Website:
http://scanme.nmap.org

Assessment Type:

Passive Vulnerability Assessment

Scope:

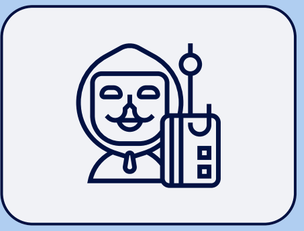
Publicly accessible pages of the website were analyzed.

Limitations:

Only passive scanning techniques were used. No intrusive or destructive tests were performed during the assessment.

Tools **used**

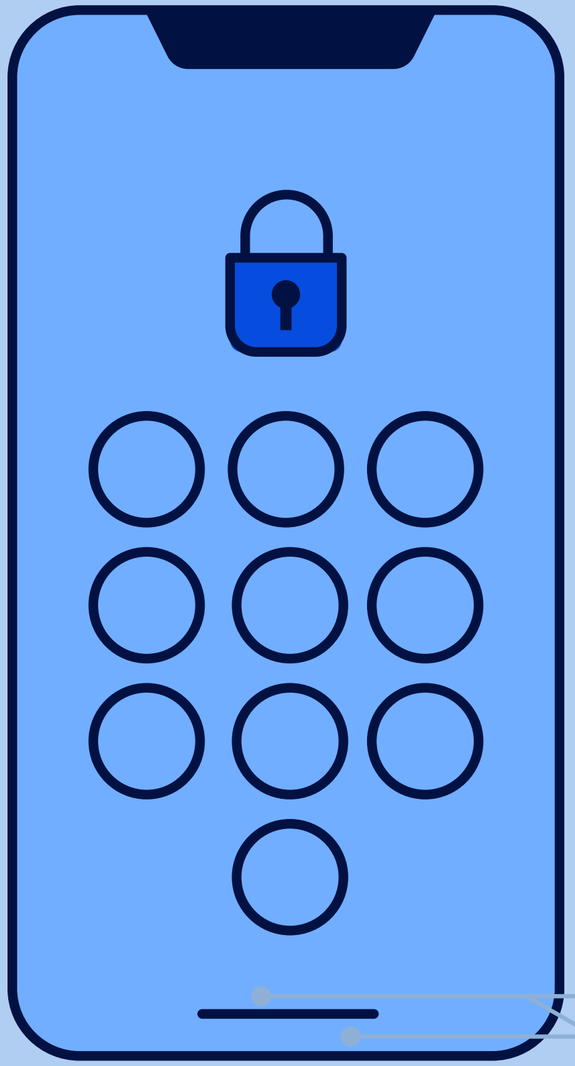
The following tools were used during the vulnerability assessment:



- **Nmap** – Network scanning and service detection
- **OWASP ZAP** – Passive vulnerability scanning
- **Browser Developer Tools** – HTTP header inspection
- **Canva** – Professional report design

Methodology **used**

Password Protected



The vulnerability assessment followed these steps:

1. Target Identification
The target website was identified and verified to ensure it was appropriate for testing.

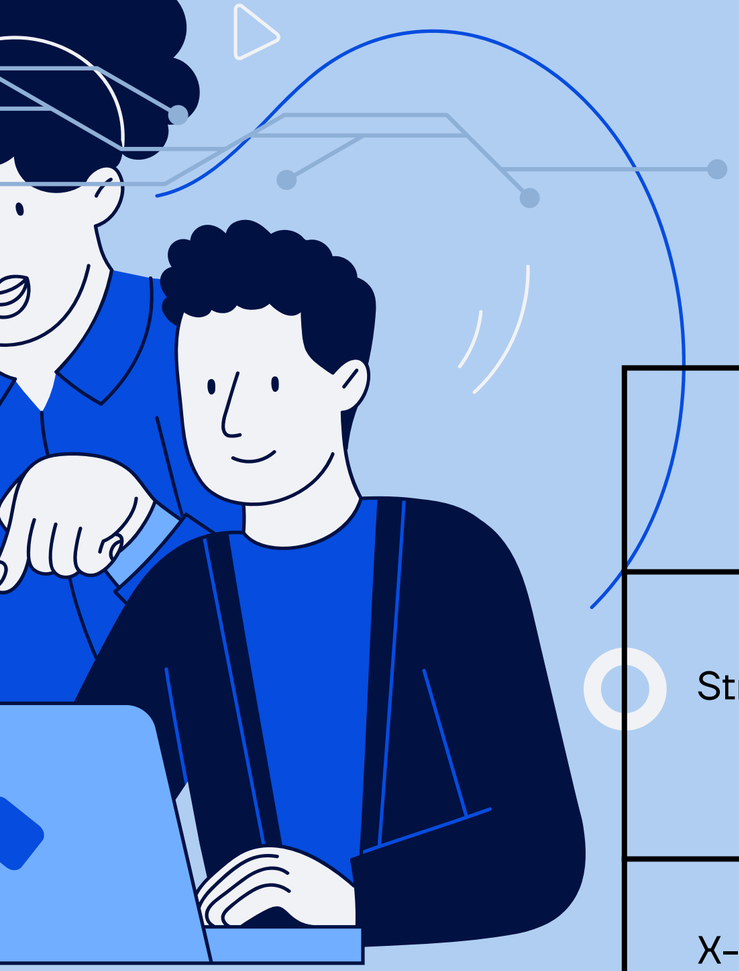
2. Network Scanning
A network scan was conducted to identify open ports and services.

3. Passive Vulnerability Scanning
OWASP ZAP was used to analyze HTTP responses and detect security misconfigurations.

.....Methodology **used**

4. Manual Inspection
Browser developer tools were used to inspect HTTP headers and response information.

5. Documentation
All findings were documented along with risk levels and remediation recommendations.

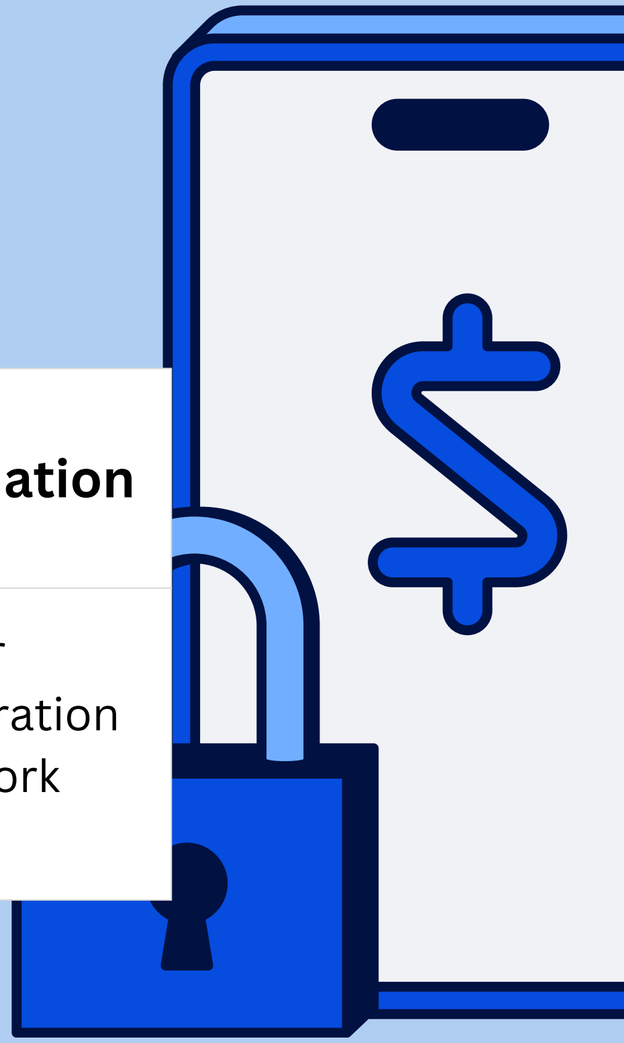


Findings of the **OWASP ZAP**

vulnerability	**Risk Level**	**Description**	**Recommendation**
Strict-Transport-Security Header Not Set	Medium	The server does not enforce HTTPS connections.	Configure HSTS header to enforce secure communication.
X-Content-Type-Options Header Missing	Low	Browser may incorrectly interpret content types.	Add X-Content-Type-Options: nosniff header.
Information Disclosure – Suspicious Comments	Low	Code comments may reveal internal information.	Remove unnecessary comments from production code.

Findings of the **Nmap**

Finding	Risk	Description	Recommendation
Firewall Filtering Detected	Low	The scan indicates that ports are filtered by firewall rules preventing detection.	Maintain proper firewall configuration to control network access.



Evidence

Nmap Scan Results

Command Prompt

```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Lenovo>nmap
Nmap 7.98 ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] {target specification}
TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iL <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3],...>: Exclude hosts/networks
  --excludefile <exclude_file>: Exclude list from file
HOST DISCOVERY:
  -sL: List Scan - simply list targets to scan
  -sn: Ping Scan - disable port scan
  -Pn: Treat all hosts as online -- skip host discovery
  -PS/PA/PU/PY[portlist]: TCP SYN, TCP ACK, UDP or SCTP discovery to given ports
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes
  -PO[protocol list]: IP Protocol Ping
  -n/-R: Never do DNS resolution/Always resolve [default: sometimes]
  --dns-servers <serv1[,serv2],...>: Specify custom DNS servers
  --system-dns: Use OS's DNS resolver
  --traceroute: Trace hop path to each host
SCAN TECHNIQUES:
  -sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
  -sU: UDP Scan
  -sN/sF/sX: TCP Null, FIN, and Xmas scans
  --scanflags <flags>: Customize TCP scan flags
  -sI <zombie host[:probeport]>: Idle scan
```

Command Prompt

```
  --unprivileged: Assume the user lacks raw socket privileges
  -V: Print version number
  -h: Print this help summary page.
EXAMPLES:
  nmap -v -A scanme.nmap.org
  nmap -v -sn 192.168.0.0/16 10.0.0.0/8
  nmap -v -iR 10000 -Pn -p 80
SEE THE MAN PAGE (https://nmap.org/book/man.html) FOR MORE OPTIONS AND EXAMPLES

C:\Users\Lenovo>nmap -sV testphp.vulnweb.com
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-14 21:20 +0400
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 5.92 seconds

C:\Users\Lenovo>
```

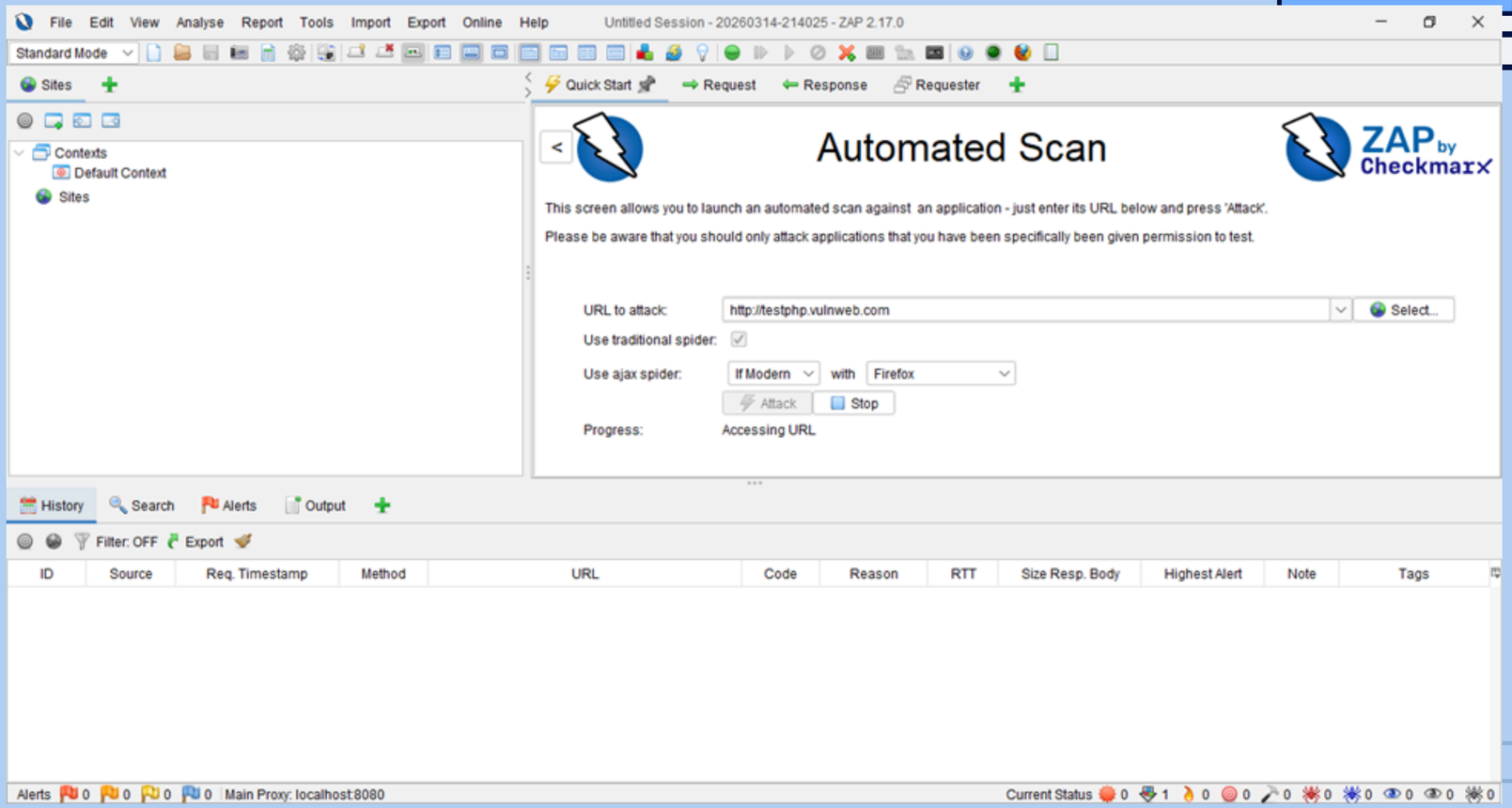
Evidence

Nmap Scan Results

```
C:\Users\Lenovo>nmap -sV -Pn testphp.vulnweb.com
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-14 21:22 +0400
Nmap scan report for testphp.vulnweb.com (44.228.249.3)
Host is up.
rDNS record for 44.228.249.3: ec2-44-228-249-3.us-west-2.compute.amazonaws.com
All 1000 scanned ports on testphp.vulnweb.com (44.228.249.3) are in ignored states
```

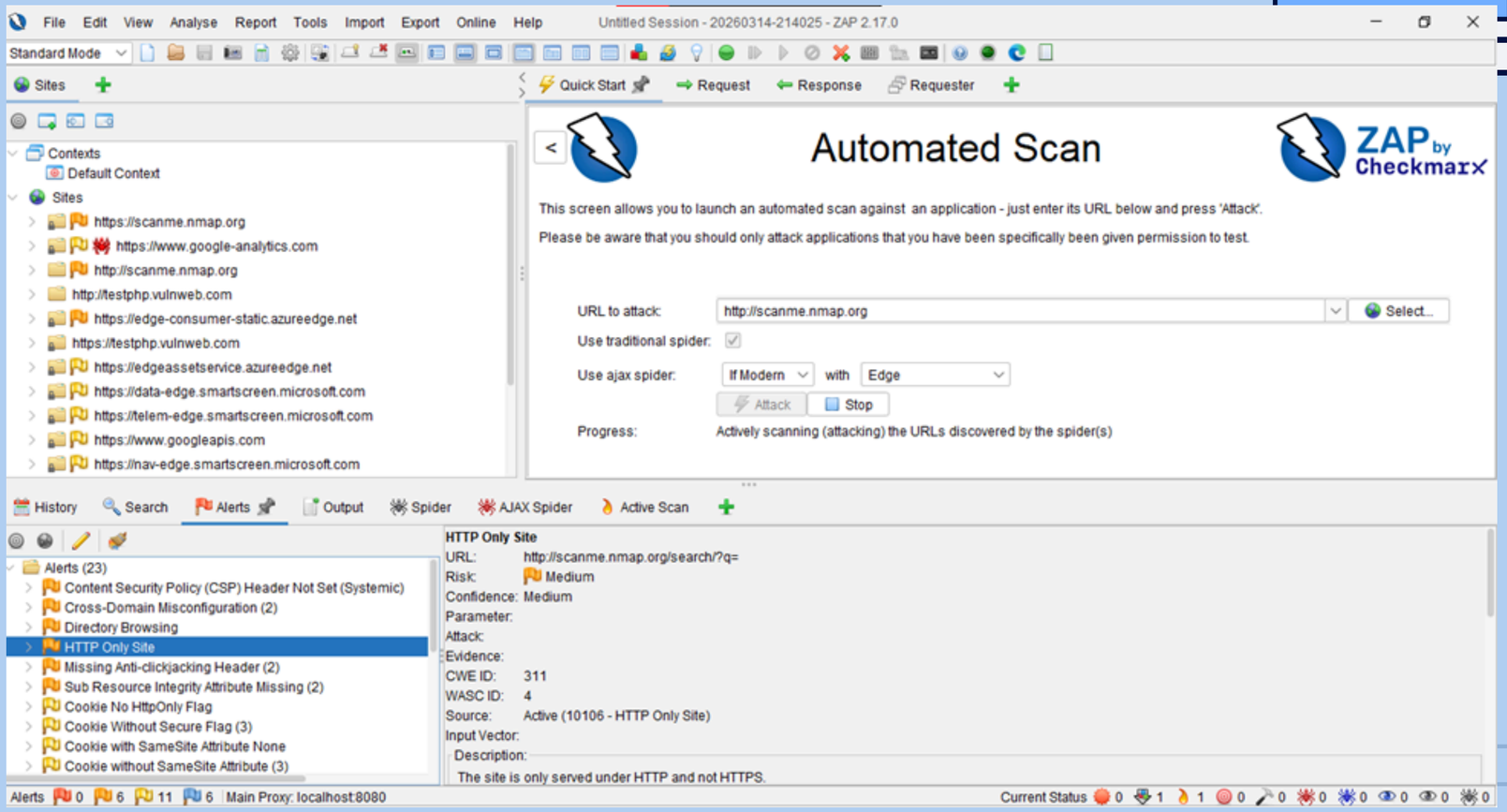
Evidence

OWASP ZAP Alerts



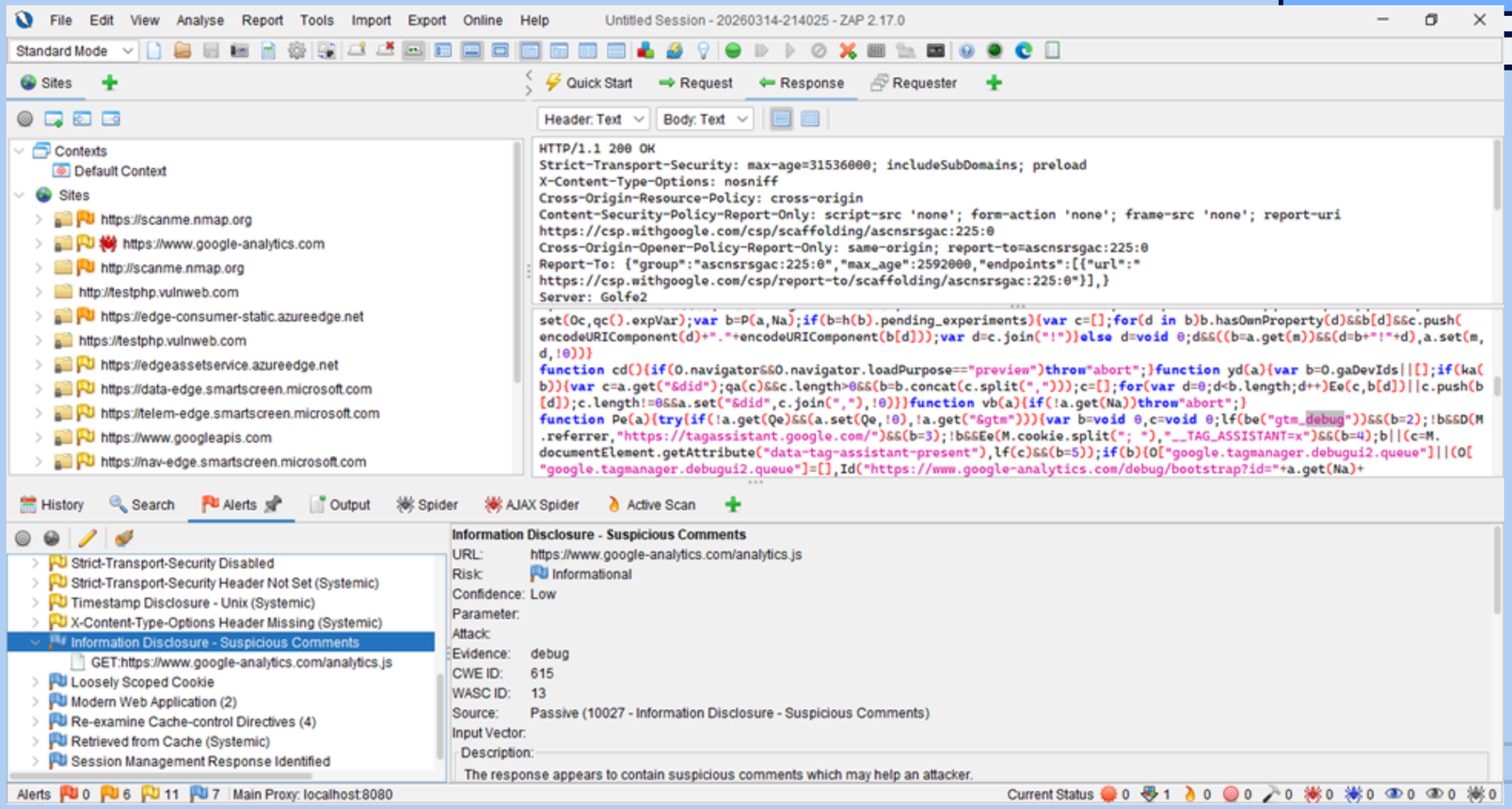
Evidence

OWASP ZAP Alerts



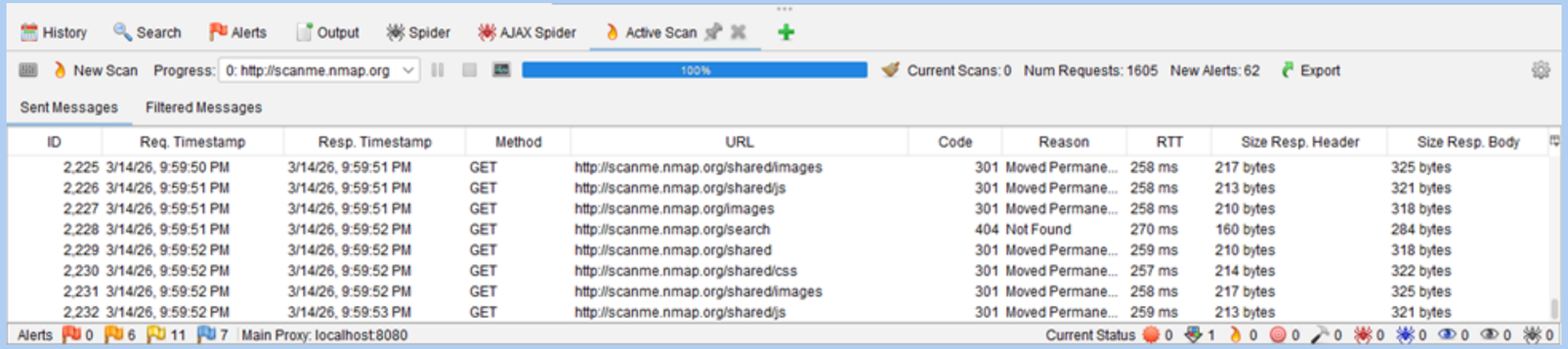
Evidence

OWASP ZAP Alerts



Evidence

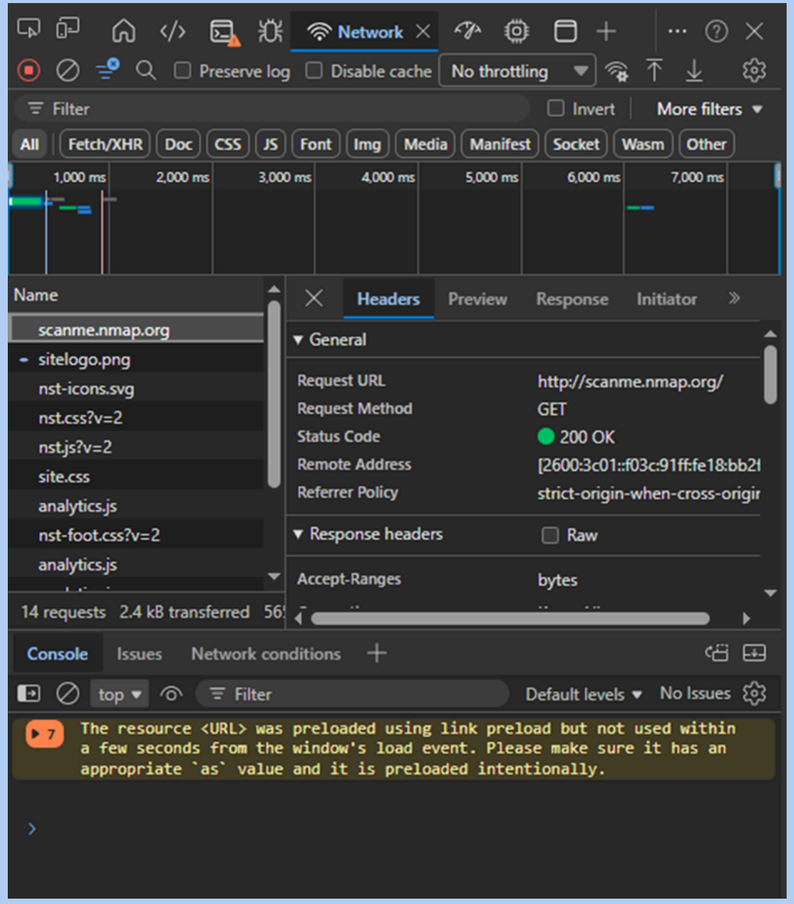
OWASP ZAP Active scans



ID	Req. Timestamp	Resp. Timestamp	Method	URL	Code	Reason	RTT	Size Resp. Header	Size Resp. Body
2,225	3/14/26, 9:59:50 PM	3/14/26, 9:59:51 PM	GET	http://scanme.nmap.org/shared/images	301	Moved Permane...	258 ms	217 bytes	325 bytes
2,226	3/14/26, 9:59:51 PM	3/14/26, 9:59:51 PM	GET	http://scanme.nmap.org/shared/js	301	Moved Permane...	258 ms	213 bytes	321 bytes
2,227	3/14/26, 9:59:51 PM	3/14/26, 9:59:51 PM	GET	http://scanme.nmap.org/images	301	Moved Permane...	258 ms	210 bytes	318 bytes
2,228	3/14/26, 9:59:51 PM	3/14/26, 9:59:52 PM	GET	http://scanme.nmap.org/search	404	Not Found	270 ms	160 bytes	284 bytes
2,229	3/14/26, 9:59:52 PM	3/14/26, 9:59:52 PM	GET	http://scanme.nmap.org/shared	301	Moved Permane...	259 ms	210 bytes	318 bytes
2,230	3/14/26, 9:59:52 PM	3/14/26, 9:59:52 PM	GET	http://scanme.nmap.org/shared/css	301	Moved Permane...	257 ms	214 bytes	322 bytes
2,231	3/14/26, 9:59:52 PM	3/14/26, 9:59:52 PM	GET	http://scanme.nmap.org/shared/images	301	Moved Permane...	258 ms	217 bytes	325 bytes
2,232	3/14/26, 9:59:52 PM	3/14/26, 9:59:53 PM	GET	http://scanme.nmap.org/shared/js	301	Moved Permane...	259 ms	213 bytes	321 bytes

Evidence

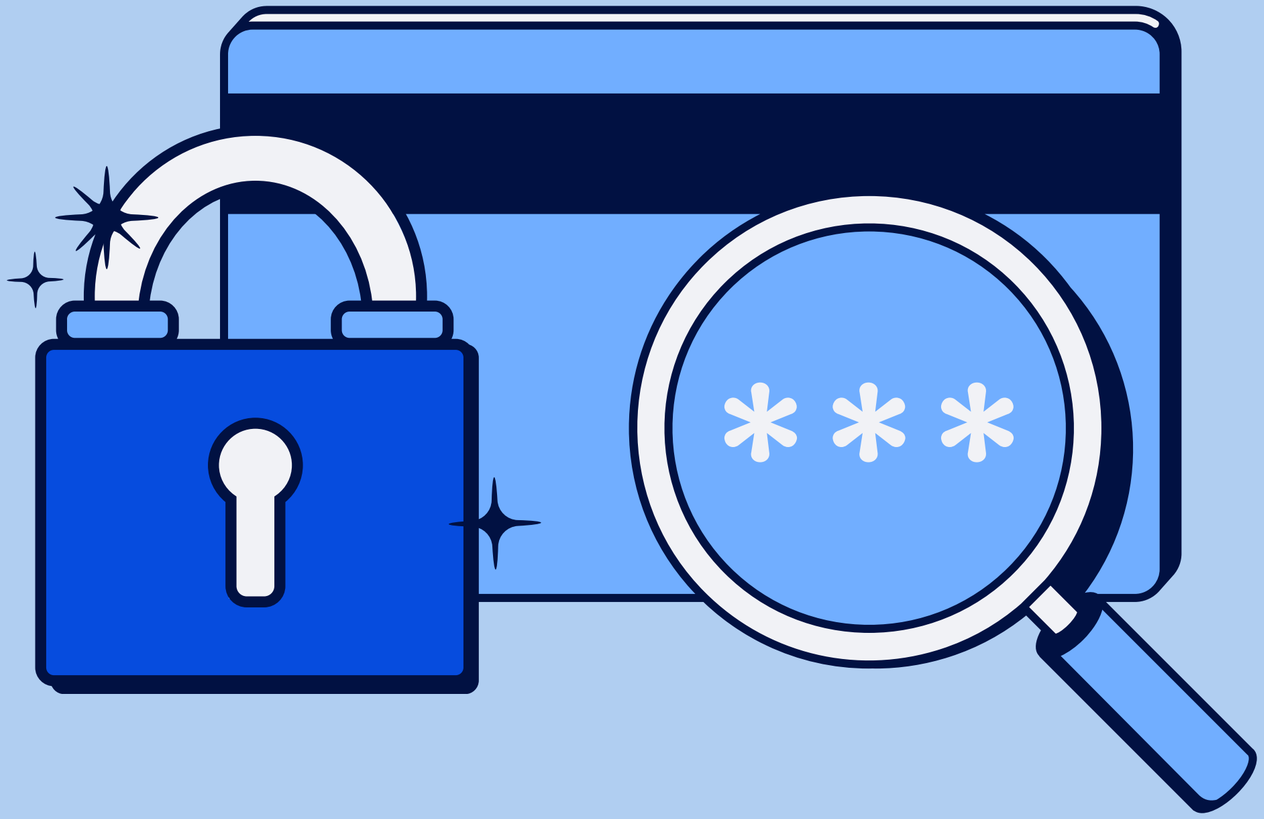
Browser Header inspection



Conclusion

The vulnerability assessment identified several low and medium level security issues related to HTTP security headers and information disclosure.

While no critical vulnerabilities were detected, implementing the recommended security configurations would significantly improve the overall security posture of the application.

Regular security testing and proper configuration management are recommended to maintain a strong security environment.

Thank You